

Shodh AI × A123 Systems

Industrial Case Study & Manufacturability Validation Portfolio

Shodh AI

May 2026

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SHODH AI × A123 SYSTEMS

Industrial Case Study & Manufacturability Validation Portfolio

Physics-native AI for battery manufacturing, process optimization, and inverse process invention

Prepared for: A123 Systems strategic technical and manufacturing evaluation

Prepared by: Shodh AI

Version: Client-ready portfolio

Date: May 2026

Classification: Confidential A123 discussion material

Basis of Presentation

All case studies are presented as normalized, NDA-compliant summaries. Detailed customer records and validation data can be reviewed under appropriate confidentiality terms.

1. Executive Summary

A123 Systems does not need another AI system that only proposes promising battery chemistry. The hard bottleneck is manufacturability.

A cell material can look exceptional in simulation or in a coin cell, then fail in the factory because of slurry agglomeration, slot-die coating instability, calendaring fracture, electrolyte wetting delay, formation bottlenecks, gas trapping, or thermal non-uniformity.

Shodh AI's thesis is:

Discovery is becoming cheap. Manufacturing remains the bottleneck. Shodh AI sits between chemistry and factory-scale production, using physics-native AI to compile manufacturable process conditions before pilot-line capital is spent.

The A123 proposal is built around this exact shift: from chemical discovery to factory-scale manufacturability. Shodh AI couples quantum, mesoscale, and factory-scale physics so new chemistries can be designed around slurry, coating, calendaring, wetting, thermal, and formation constraints before expensive pilot iteration.

This portfolio shows how Shodh's physics model has two commercial modes:

1. **Optimization Mode** — rescue or improve an existing process by identifying hidden physical bottlenecks and generating better operating policies.
2. **Invention Mode** — generate new process hardware, reactor geometries, material architectures, or operating protocols that human teams would normally search for through years of physical experimentation.

For A123, these translate directly into:

- lower scrap rates,
 - faster process windows,
 - shorter electrolyte wetting time,
 - fewer failed pilot builds,
 - better silicon and semi-solid manufacturability,
 - more robust thermal management,
 - patentable process and structural IP,
 - faster transition from material concept to production cell.
-

2. Why These Case Studies Matter for A123

The case studies are not presented because pharma and specialty chemicals are identical to batteries. They are presented because the **underlying scale-up physics is analogous**.

Cross-Industry Case	What Shodh Demonstrated	A123 Translation
Aarti Industries	Optimized an exothermic, mixing-limited process where hotspots and dead zones destroyed yield	Slurry mixing, thermal gradients, coating uniformity, and factory defect localization
Jubilant Pharmova	Generated a new continuous-flow process architecture under multiphase transport constraints	Ultra-wetting electrode architecture, immersion-cooling geometry, and process hardware invention
Biocon / biologics analog	Managed shear-sensitive living systems during scale-up	Silicon particle fracture, binder-network damage, and shear-sensitive semi-solid matrices
Agrochemical solvent analog	Inverse-designed a formulation against viscosity, freezing-point, and regulatory constraints	Electrolyte, binder, gel, and semi-solid formulation design under manufacturability constraints

The unifying concept is not industry-specific chemistry. It is the ability to model and optimize coupled physics across scale.

Batteries fail during scale-up because multiple domains interact:

- chemistry,
- rheology,
- porous media,
- capillary wetting,
- solid mechanics,
- electrochemistry,
- heat transfer,
- fluid dynamics,
- factory control.

Shodh AI is built to route these domains into specialized physics experts while preserving cross-scale coupling.

3. Shodh AI Platform Overview

3.1 Input Layer

A deployment may begin with:

- equipment CAD or GA drawings,
- SCADA / PLC logs,
- mixing speed, pressure, temperature, torque, and flow-rate histories,
- slurry solids loading,
- relative viscosity curves,
- slot-die coating parameters,
- calendaring pressure and roll-gap data,
- electrolyte wetting time,
- formation-cycle data,
- QC pass/fail outcomes,
- scrap category labels,
- cell impedance, capacity, and cycle-life measurements.

A123 can obfuscate chemistry names and normalize telemetry. The model learns from physical gradients, time

evolution, constraints, and correlations rather than requiring raw proprietary formula labels.

3.2 Model Layer

Shodh AI combines:

- physics-constrained neural operators,
- multiphysics routing experts,
- geometry-aware field tokenization,
- differentiable simulation,
- inverse adjoint optimization,
- factory-specific adapter learning,
- manufacturability screening under strict PDE constraints.

3.3 Output Layer

The model returns:

- defect localization maps,
- process-window recommendations,
- RPM curves,
- pressure thresholds,
- coating speed windows,
- calendaring pressure bands,
- wetting protocols,
- formation-current schedules,
- thermal geometry suggestions,
- generated CAD / SDF geometries,
- ranked validation experiments,
- projected ROI under stated assumptions.

3.4 Deployment Principle

The goal is not to eliminate physical validation. The goal is to reduce the number of physical experiments needed before confidence is high.

4. A123-Specific Pilot Architecture

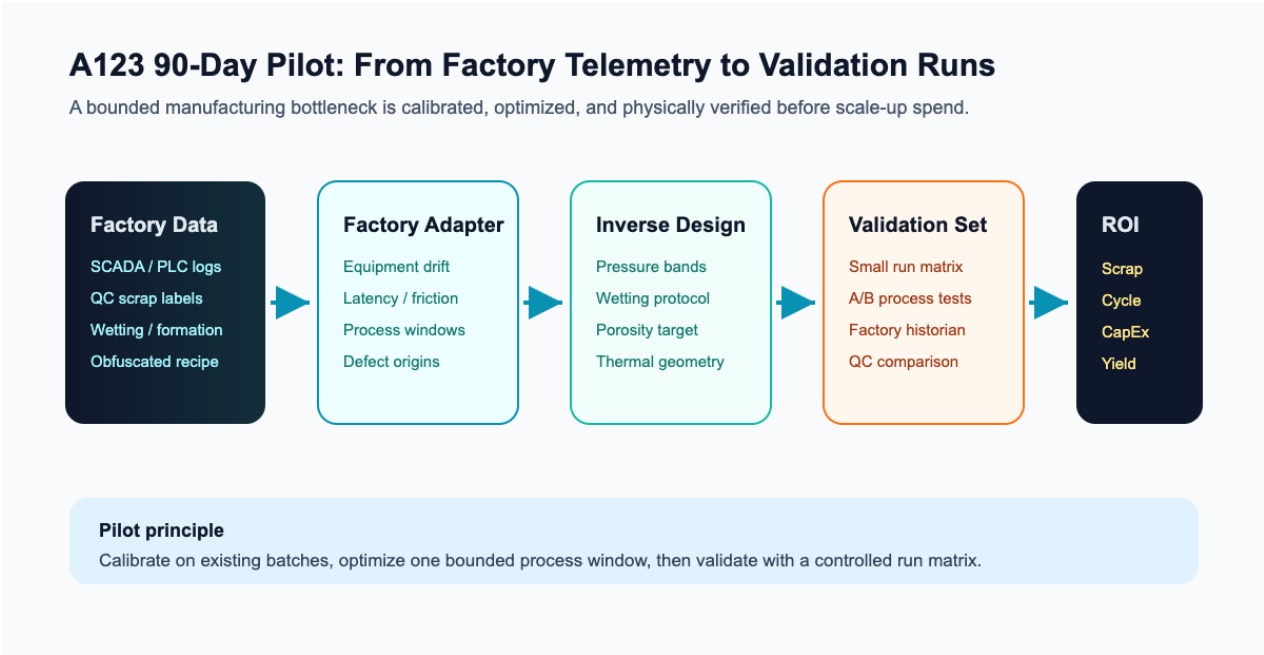


Figure 1: A123 90-day pilot workflow: Phase 1 calibrates a factory adapter from historical telemetry, and Phase 2 converts that adapter into inverse process recommendations, validation runs, and ROI measurement.

4.1 Phase 1 — Factory Calibration

Objective: prove predictive value on A123’s existing production data before asking the client to trust Shodh with future materials or new cell architectures.

A123 selects one bounded bottleneck:

- slurry mixing,
- slot-die coating,
- calendering,
- electrolyte wetting,
- formation,
- thermal management.

Shodh ingests approximately 20 historical batches, including both high-performing and scrapped batches. For each batch, A123 provides obfuscated process telemetry and final QC outcome.

The model reconstructs each batch and predicts the physical origin of defects.

Phase 1 Deliverables

Deliverable	What A123 Receives	Commercial Relevance
Digital factory diagnostic	Spatial and temporal maps of likely defect origins	Shows where scrap is created, not just where it is detected
Factory Adapter	Lightweight calibrated model layer representing A123-specific friction, drift, latency, and equipment behavior	Future simulations reflect A123’s real factory, not idealized textbook physics
Process-window recommendations	RPM, flow, pressure, temperature, wetting, or coating windows	Immediate scrap reduction opportunity
Validation experiment set	Small number of recommended confirmatory runs	Reduces blind process experiments

4.2 Phase 2 — Joint Manufacturing IP Co-Development

After calibration, Shodh moves from diagnosing the past to engineering future A123 manufacturing and material IP.

A123 may select one or more tracks:

Track A — Expansion-Tolerant Semi-Solid Matrix

Bottleneck: silicon anodes can expand dramatically during cycling, causing tearing, delamination, and capacity fade.

Shodh output: target viscoelasticity, cross-link density, polymer composition bands, and expansion-tolerant matrix architecture.

IP outcome: composition-of-matter or formulation patent.

Track B — Ultra-Wetting Electrode Architecture

Bottleneck: high-viscosity gels and dense electrodes create capillary wetting delays and trapped gas pockets.

Shodh output: calendering porosity, pore-size distribution, laser microchannel architecture, and wetting protocol.

IP outcome: structural / utility patent and faster finishing throughput.

Track C — Silicon-Protective Calendering + Flash Formation Protocol

Bottleneck: excessive calendering pressure fractures high-silicon structures, while slow formation ties up floor space and working capital.

Shodh output: pressure band, roll-gap policy, microscopic buffer-void target, and pulsed-current formation protocol.

IP outcome: process patents and faster line qualification.

Track D — Immersion-Cooling Flow Geometry Optimization

Bottleneck: thick dielectric cooling fluids can create pressure drop and parasitic pumping loss while failing to maximize heat transfer.

Shodh output: module exterior geometry, micro-turbulence features, pressure-drop maps, and heat-transfer optimized casing structures.

IP outcome: mechanical system patent and improved thermal runaway mitigation.

5. Battery Manufacturing Governing Physics

This section gives A123's technical reviewers confidence that Shodh is not offering a generic AI workflow. Each manufacturing defect class is tied to governing equations and measurable factory variables.

5.1 Slurry Mixing and Agglomeration

Battery slurry behavior is non-Newtonian. A Herschel-Bulkley form captures yield stress and shear-thinning behavior:

$$\tau = \tau_y + K\dot{\gamma}^n$$

where τ_y is yield stress, K is consistency index, $\dot{\gamma}$ is shear rate, and $n < 1$ indicates shear thinning.

Agglomeration risk increases when local shear is too low to break clusters or too high and damages binder networks:

$$\Pi_{break} = \frac{\tau_{local}}{\sigma_{agg}}$$

A useful operating window is:

$$\Pi_{min} < \Pi_{break} < \Pi_{damage}$$

This means the process must create enough local stress to disperse agglomerates without damaging particles, binder, or semi-solid networks.

5.2 Slot-Die Coating Stability

Wet-film thickness is constrained by volumetric flow rate, web speed, and coating width:

$$h_w = \frac{Q}{UW}$$

where Q is flow rate, U is web speed, and W is coating width.

The capillary number governs the balance between viscous and surface-tension forces:

$$Ca = \frac{\mu U}{\gamma}$$

Coating defects emerge when Ca , viscosity, lip geometry, and pressure fluctuations leave the stable coating window.

5.3 Calendering and Particle Fracture

Calendering must reduce porosity and improve contact without crushing active material.

A representative compaction model:

$$\epsilon(P) = \epsilon_0 e^{-\alpha P}$$

where ϵ is porosity, P is applied pressure, and α is material-specific compaction response.

Fracture risk can be stated as:

$$\sigma_{vM}(P, \epsilon, d_p) < \sigma_{fracture}$$

where σ_{vM} is von Mises stress, d_p is particle size, and $\sigma_{fracture}$ is the fracture threshold.

5.4 Electrolyte Wetting

Capillary wetting through porous electrodes follows a Washburn-style scaling:

$$L^2 = \frac{r_p \gamma \cos \theta}{2\mu} t$$

where L is penetration depth, r_p is effective pore radius, γ is surface tension, θ is contact angle, μ is viscosity, and t is time.

A123's wetting bottleneck can be attacked through:

- pore architecture,
- calendering pressure,
- surface energy,
- electrolyte viscosity,
- vacuum / pressure cycling,
- microchannel architecture.

5.5 Formation and SEI Growth

Formation kinetics are governed by electrochemical reaction rate, transport, and thermal behavior.

A representative Butler-Volmer equation:

$$i = i_0 \left[\exp \left(\frac{\alpha_a F \eta}{RT} \right) - \exp \left(- \frac{\alpha_c F \eta}{RT} \right) \right]$$

SEI growth can be modeled as:

$$\frac{d\delta_{SEI}}{dt} = k_{SEI} \exp \left(- \frac{E_a}{RT} \right) f(\eta, c_{Li^+})$$

The optimization goal is to accelerate formation without creating lithium plating, gas evolution, thermal excursions, or impedance penalties.

5.6 Thermal Management

Cell and module heat transfer is governed by:

$$\rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) = k \nabla^2 T + Q_{ohmic} + Q_{rxn}$$

For immersion cooling, the design objective is to increase heat-transfer coefficient while controlling pumping loss:

$$\max(hA) \quad \text{subject to} \quad \Delta P < \Delta P_{max}$$

This is directly analogous to Shodh’s inverse geometry work in reactor and flow systems.

6. Portfolio Snapshot

Case	Client / Archetype	Paradigm	Primary Physics	Evidence Basis	Value Claim Format
1	Aarti Industries	Brownfield optimization	Exothermic nitration, turbulence, heat transfer	NDA-compliant deployment summary	Yield, impurity, COGS, time-to-market
2	Jubilant Pharmova	Process / hardware invention	Multiphase catalysis, phase-field interfaces, surface kinetics	NDA-compliant deployment summary	Residence time, purity, CapEx avoidance
3	Biologics analog	Scale-up risk removal	Shear, eddy dissipation, oxygen transfer	Cross-industry analog	Pilot batches, titer, time-to-market
4	Agrochemical solvent analog	De novo formulation	DFT, solvent thermodynam- ics, viscosity, freezing point	Cross-industry analog	Discovery time, wet-lab burden, regulatory continuity
5	A123 Systems proposed pilot	Battery manu- facturability	Slurry, coating, wetting, calendering, formation	Proposed A123 pilot success model	Scrap reduction, wetting time, U/hr, validation accuracy

7. Case Study 1 — Aarti Industries

Brownfield Optimization: Exothermic Nitration Scale-Up

Case purpose for A123: prove that Shodh can identify hidden physical bottlenecks inside active industrial equipment and convert them into process-window improvements.

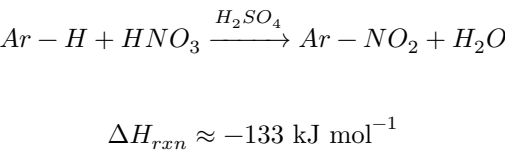
A123 analogy: slurry mixer dead zones, coating instability, thermal non-uniformity, and defect localization.

Evidence basis: NDA-compliant deployment summary. Quantitative values are presented as normalized customer-impact metrics.

7.1 Industrial Problem

Aarti-style specialty chemical production often fails during the transition from lab or pilot scale to commercial equipment. In the case pattern, an exothermic nitration of a substituted aromatic compound is scaled into continuous or large stirred equipment.

Representative chemistry:



At small scale, heat removal is easy. At plant scale, localized fluid zones produce high residence time and thermal pooling. Because nitration kinetics are temperature-sensitive, small hotspots accelerate secondary nitration and create costly impurities.

The failure mode is not a chemistry failure. It is a manufacturing physics failure.

7.2 Counterfactual Without Shodh

- The traditional path requires:
1. intermediate pilot hardware,
 2. iterative CFD runs,
 3. operator-driven recipe tuning,
 4. experimental changes in feed location and mixing speed,
 5. repeated impurity analysis,
 6. final process safety review.

Traditional counterfactual:

Counterfactual Item	Traditional Path
Intermediate pilot iterations	3 major hardware / recipe loops
Development time	18–24 months
Engineering + pilot CapEx	\$3.0M–\$14.0M depending on equipment scope
Baseline commercial yield	84.5%
Major defect	secondary di-nitration and thermal runaway risk

7.3 Governing Physics

Flow Field

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}$$

Species Transport

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (D_i \nabla Y_i) + R_i$$

Arrhenius Kinetics

$$r = k_0 \exp\left(-\frac{E_a}{RT}\right) [A][B]$$

Energy Balance

$$\rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) = k \nabla^2 T + (-\Delta H_{rxn}) r$$

Mixing-Reaction Coupling

$$Da = \frac{\tau_{mix}}{\tau_{reaction}}$$

If $Da \gg 1$, reaction occurs faster than mixing, increasing local impurity formation. The goal is to drive the effective process toward a controlled $Da \approx O(1)$ window.

7.4 Shodh AI Intervention

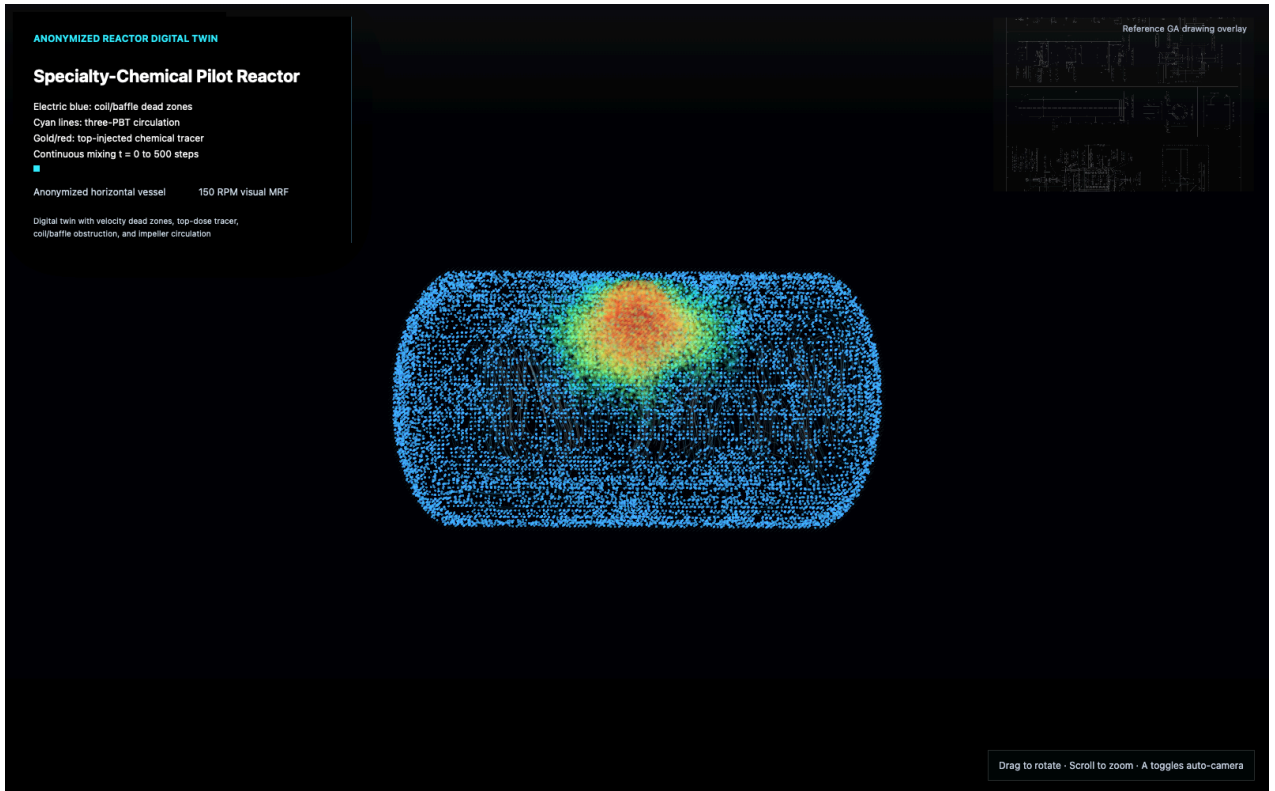


Figure 2: Anonymized exothermic nitration reactor digital twin showing top-dose tracer mixing, coil / baffle dead zones, and impeller circulation.

Shodh ingested:

- reactor or mixer geometry,
- feed conditions,
- operating histories,
- reaction kinetics,
- temperature and impurity data,
- target yield and thermal constraints.

The model diagnosed a high-residence-time zone under the mixing structure and near the feed interaction region. This zone created a local temperature excursion and excessive secondary reaction probability.

The inverse adjoint loop backpropagated yield and thermal variance error into process recommendations:

- feed-port angle change,

- injection pressure adjustment,
- dynamic RPM curve,
- cooling-jacket temperature gradient,
- static mixer or impeller geometry candidate.

Optimization objective:

$$\mathcal{L} = -Y_{primary} + \lambda_1 \sigma_T^2 + \lambda_2 I_{secondary} + \lambda_3 \Delta P + \lambda_4 E_{cooling}$$

where $Y_{primary}$ is desired product yield, σ_T^2 is thermal variance, $I_{secondary}$ is impurity formation, ΔP is pressure drop, and $E_{cooling}$ is utility cost.

7.5 AI-Generated Operating Recipe

Process recommendation summary:

Parameter	Baseline	Shodh AI Recommendation	Reason
Feed rate	38.0 L/min	42.5 L/min	Increased throughput after eliminating hotspot risk
Feed angle	90° radial	37° tangential-downward	Better mixing before high-temperature zone
Agitation	fixed 400 RPM	220–760 RPM pulsed curve	Breaks stagnant structures without over-shearing
Jacket profile	constant 25°C	15°C inlet to 45°C outlet gradient	Matches heat release profile
Static insert	none	fractal / helical insert candidate	Reduces residence-time variance

7.6 Validation Summary

Validation Chart 1 — Yield by Run

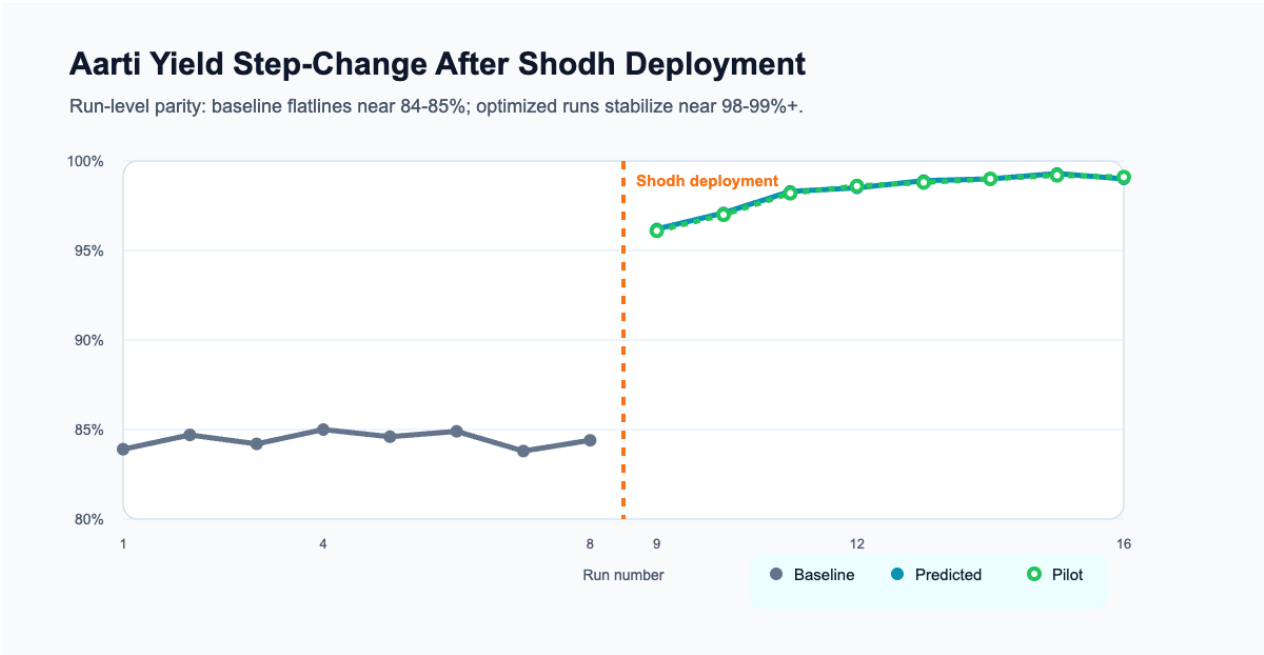


Figure 3: Specialty-chemical yield step-change after Shodh deployment, showing predicted and pilot parity against the baseline process.

Key takeaway: baseline runs remain stuck near 84–85% yield; optimized runs stabilize near 98–99%+, with predicted and pilot values closely aligned.

Validation Chart 2 — Impurity Reduction and Process Stability



Figure 4: Specialty-chemical impurity and process-stability improvement after Shodh optimization.

Metric	Baseline Mean	Shodh Mean	Improvement
Mono-nitrated yield	84–85%	98–99%	+13–15 percentage points
Di-nitrated impurity	4–5%	<0.1%	>95% reduction
Thermal variance	18.4 °C ²	3.1 °C ²	83.2% reduction
Effective Damköhler number	3.20	1.08	moved into controlled regime
Yield prediction error	—	<1 percentage point	sim-to-real parity target

Validation Chart 3 — COGS Bridge



Figure 5: Specialty-chemical COGS bridge by cost component, separating raw material loss, solvent recovery, waste disposal, energy, and QA burden.

Cost Component	Baseline Index	Shodh Index	Change
Raw material loss	100	86.6	-13.4
Solvent recovery	100	21.7	-78.3
Waste disposal	100	12.5	-87.5
Energy / cooling	100	67.9	-32.1
QA / rework burden	100	12.1	-87.9
Total COGS	100	68	32% reduction

7.7 A123 Relevance

Aarti is the strongest proof that Shodh can perform **brownfield optimization**.

For A123, the analogous target is not nitration yield. It is:

- slurry uniformity,
- agglomeration suppression,
- coating defect reduction,
- fewer scrapped batches,
- more stable wetting and formation behavior,
- higher units-per-hour yield.

The same platform logic applies: ingest real factory histories, reconstruct hidden physical failure zones, calibrate a factory adapter, and recommend process changes before expensive production trials.

8. Case Study 2 — Jubilant Pharmova

Process / Hardware Invention: Continuous-Flow Multiphase Catalytic Manufacturing

Case purpose for A123: prove that Shodh can generate new process hardware and operating policies, not merely tune existing recipes.

A123 analogy: ultra-wetting electrode architecture, calendaring microstructure design, immersion-cooling casing geometry, and formation protocols.

Evidence basis: NDA-compliant deployment summary. Quantitative values are presented as normalized customer-impact metrics.

8.1 Industrial Problem

A Jubilant-style API intermediate required a high-value catalytic transformation constrained by gas-liquid-solid mass transfer and selectivity. The traditional process relied on long batch residence time or conventional packed-bed hardware.

The bottlenecks:

- poor gas-liquid contact,
- catalyst underutilization,
- pressure drop,
- gas channeling,
- long residence time,
- purity / ee loss,
- high CapEx if scaled through larger batch assets.

The key challenge was not simply catalyst chemistry. It was the design of a process architecture where chemistry, transport, pressure, and selectivity were jointly controlled.

8.2 Counterfactual Without Shodh

Development Item	Traditional Path
Reactor options tested	batch, packed bed, microchannel, static mixer
Development time	3–4 years
Planned CapEx route	\$22M batch expansion
Baseline residence time	18 hours
Baseline enantiomeric purity	64% ee
Major failure mode	gas channeling and mass-transfer limitation

8.3 Governing Physics

Phase-Field Interface Dynamics

$$\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = \nabla \cdot \left(M \nabla \left(\frac{\partial f}{\partial c} - \kappa \nabla^2 c \right) \right)$$

where c represents phase fraction, M mobility, f free energy density, and κ interfacial penalty.

Surface Catalysis

$$r_{surface} = \frac{k K_H P_H K_S [S]}{(1 + K_H P_H + K_S [S])^2}$$

where P_H is gas partial pressure, $[S]$ substrate concentration, and K_H , K_S adsorption constants.

Geometry Optimization Objective

$$\max_{\Omega, \theta} (A_{interface}, X, ee)$$

subject to:

$$\Delta P < 2 \text{ bar}, \quad \tau_{\text{residence}} < 20 \text{ min}, \quad ee > 99\%, \quad Q > Q_{\min}$$

where Ω is geometry and θ is operating policy.

8.4 Shodh AI Intervention

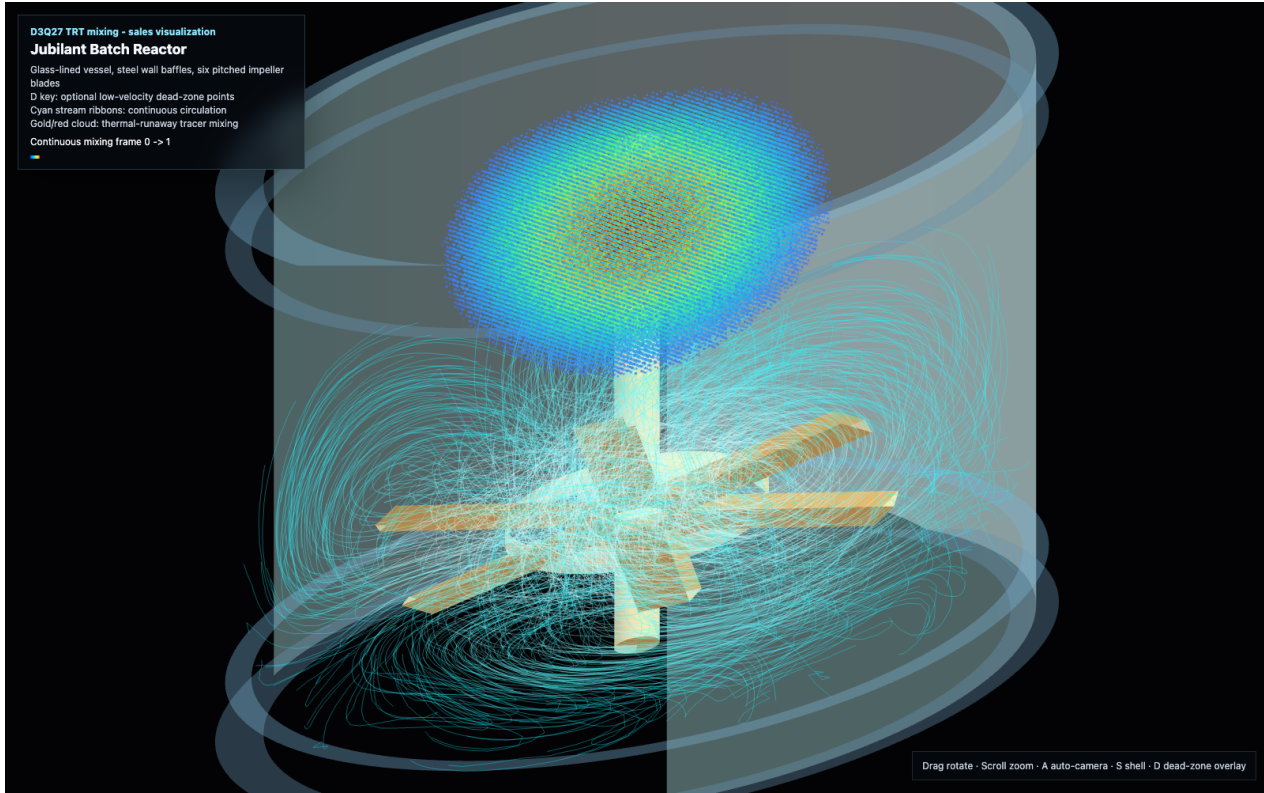


Figure 6: Baseline reactor digital twin used to identify multiphase transport limitations prior to inverse geometry generation.

Shodh initialized a manufacturable geometry search under process constraints. Rather than asking engineers to manually test one reactor at a time, the model searched a large candidate space of internal structures.

Candidate geometries included:

- packed bed variants,
- microchannel arrays,
- static mixers,
- serpentine channels,
- TPMS / gyroid lattice structures.

Shodh AI generated a TPMS-like geometry that increased usable interfacial area while maintaining a low-pressure path. The geometry behaved like a continuous-flow process intensifier: high contact area without the pressure penalty of a conventional packed bed.

8.5 AI-Generated Process Architecture

Design Variable	Baseline	Shodh Candidate	Reason
Reactor mode	batch / packed bed	continuous-flow TPMS insert	improves contact and control
Residence time	18 hours	<20 minutes	process intensification
Pressure drop	unstable / high	1.6 bar	below 2 bar constraint

Design Variable	Baseline	Shodh Candidate	Reason
Gas-liquid interface	low effective area	4.1× baseline	improves mass transfer
Catalyst utilization	channeling-limited	uniform wetting/contact	reduces dead zones
Purity / ee	64% ee	>99% ee	pharma-grade target

8.6 Validation Summary

Validation Chart J1 — Pareto Frontier

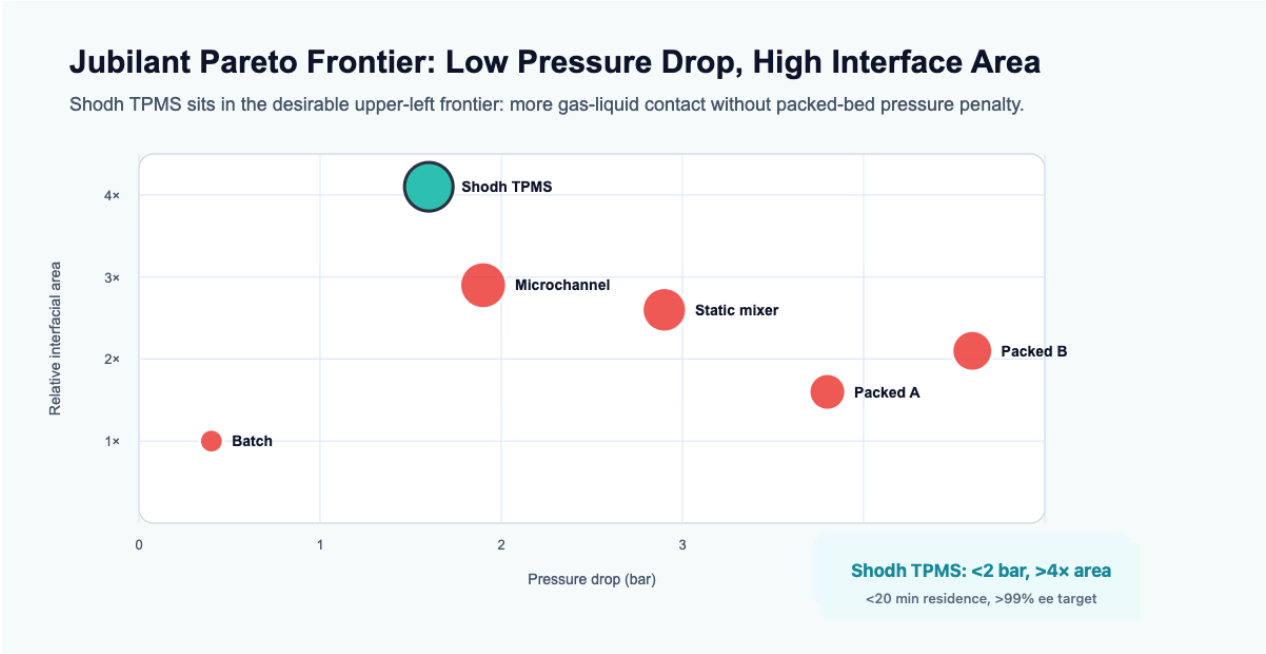


Figure 7: Jubilant reactor Pareto frontier: Shodh TPMS occupies the low-pressure, high-interface-area region.

Key takeaway: the Shodh TPMS candidate occupies the desired frontier: high interfacial area and low pressure drop, with a 14-minute target residence time.

Validation Chart J2 — Time-to-Market Compression

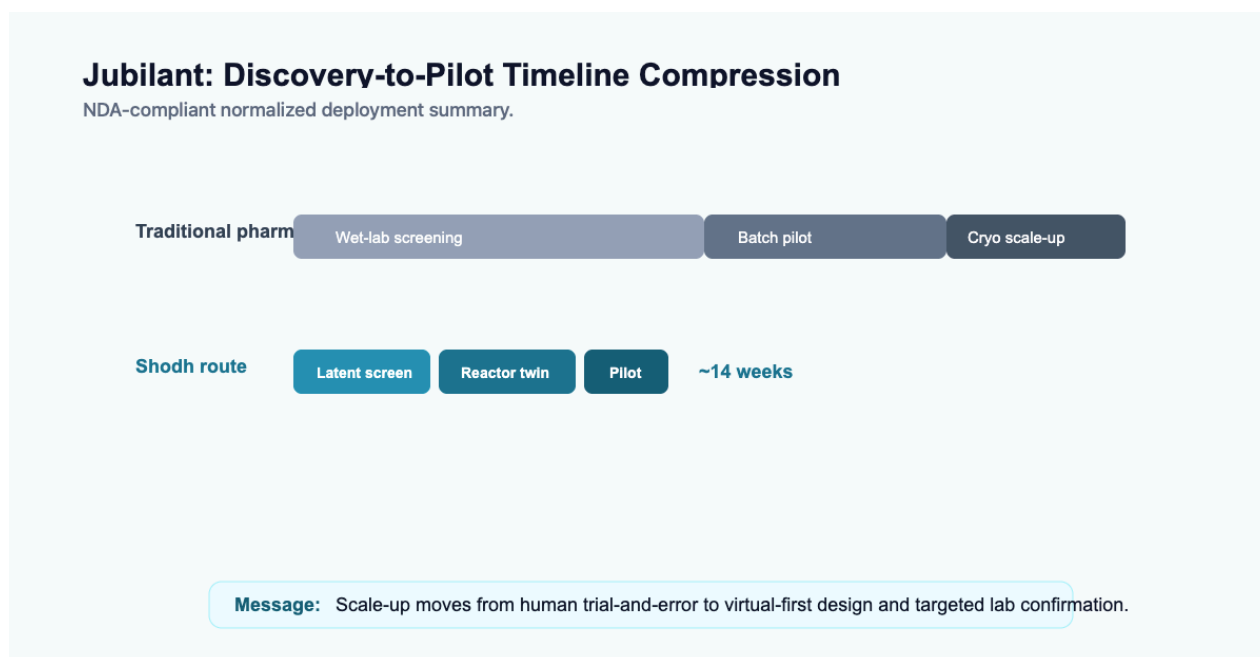


Figure 8: Jubilant time-to-market compression: wet-lab discovery and scale-up path versus Shodh-assisted virtual process invention.

Key takeaway: the process-invention path compresses a multi-year discovery and scale-up loop into targeted validation cycles measured in weeks.

Validation Chart J3 — CapEx Avoidance

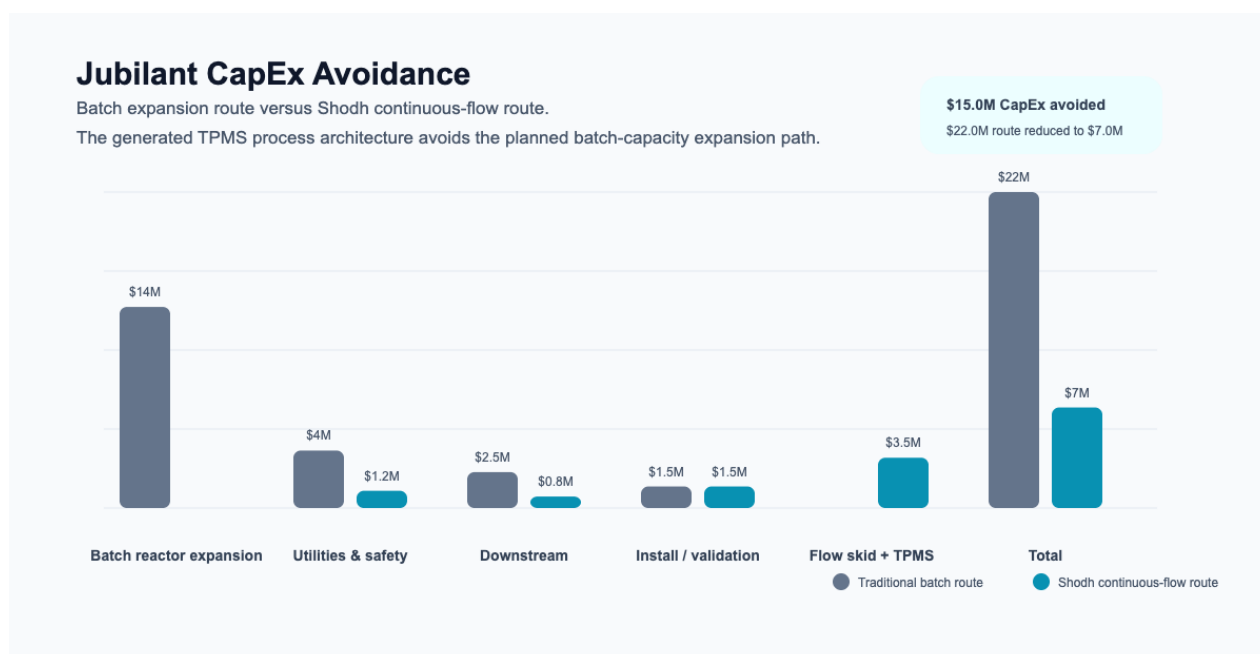


Figure 9: Jubilant CapEx avoidance: batch expansion route versus Shodh continuous-flow route.

Key takeaway: the continuous-flow route replaces a planned batch-capacity expansion with a smaller flow skid and printed/reactor-insert hardware.

8.7 A123 Relevance

Jubilant is the strongest proof of **process and hardware invention**.

For A123, the analogs are:

- ultra-wetting electrode channels,
- generated calendering porosity fields,
- immersion-cooling geometries,
- low-pressure thermal-flow structures,
- process protocols for faster formation and wetting.

The point is not that A123 needs pharma reactors. The point is that Shodh can generate manufacturable geometry under coupled physics constraints.

9. Replicability Case 3 — Biologics Scale-Up Analog

Shear-Sensitive Process Optimization

Case purpose for A123: demonstrate control of delicate structures under scale-up shear.

A123 analogy: silicon particle fracture, binder-network damage, semi-solid matrix tearing, and slurry shear sensitivity.

Evidence basis: Cross-industry analog.

9.1 Problem

CHO cells and biologic cultures are damaged when local eddy structures become comparable to cell dimensions. Similarly, advanced battery materials can be damaged by local shear or compressive stress during mixing and calendering.

The governing shear scale is the Kolmogorov length:

$$\eta_K = \left(\frac{\nu^3}{\epsilon}\right)^{1/4}$$

Cell or particle damage risk rises when:

$$\eta_K \lesssim d_{critical}$$

9.2 Representative Metrics

Metric	Traditional Path	Shodh-Assisted Path	Impact
Pilot batches	16	3	81% reduction
Scale-up R&D cost	\$8.0M	\$1.5M	\$6.5M saved
Final titer / productivity	4.2 g/L	5.1 g/L	21% improvement
Time-to-market	22 months	6 months	16 months saved

9.3 A123 Translation

Replace “cell viability” with “particle and electrode-network viability.”

A123-specific quantities:

- silicon fracture probability,
- binder-network strain,
- agglomerate breakage,
- slurry viscosity drift,
- electrode crack density after calendering,
- impedance rise after formation.

The model can search for the operating window where mixing is strong enough to disperse but not strong enough to destroy.

10. Replicability Case 4 — Agrochemical Solvent Replacement Analog

Inverse Formulation Under Viscosity, Freezing Point, and Regulatory Constraints

Case purpose for A123: demonstrate de novo formulation design under multi-property constraints.

A123 analogy: electrolyte, binder, gel, and semi-solid matrix formulation.

Evidence basis: Cross-industry analog.

10.1 Problem

A solvent replacement problem requires simultaneous optimization of:

- freezing point,
- viscosity,
- solubility,
- toxicity,
- biodegradability,
- manufacturability,
- factory mixing behavior.

Battery materials face the same multi-property constraint problem. A gel electrolyte or binder system must not only work electrochemically; it must mix, coat, wet, dry, form, and cycle.

10.2 Representative Inverse Objective

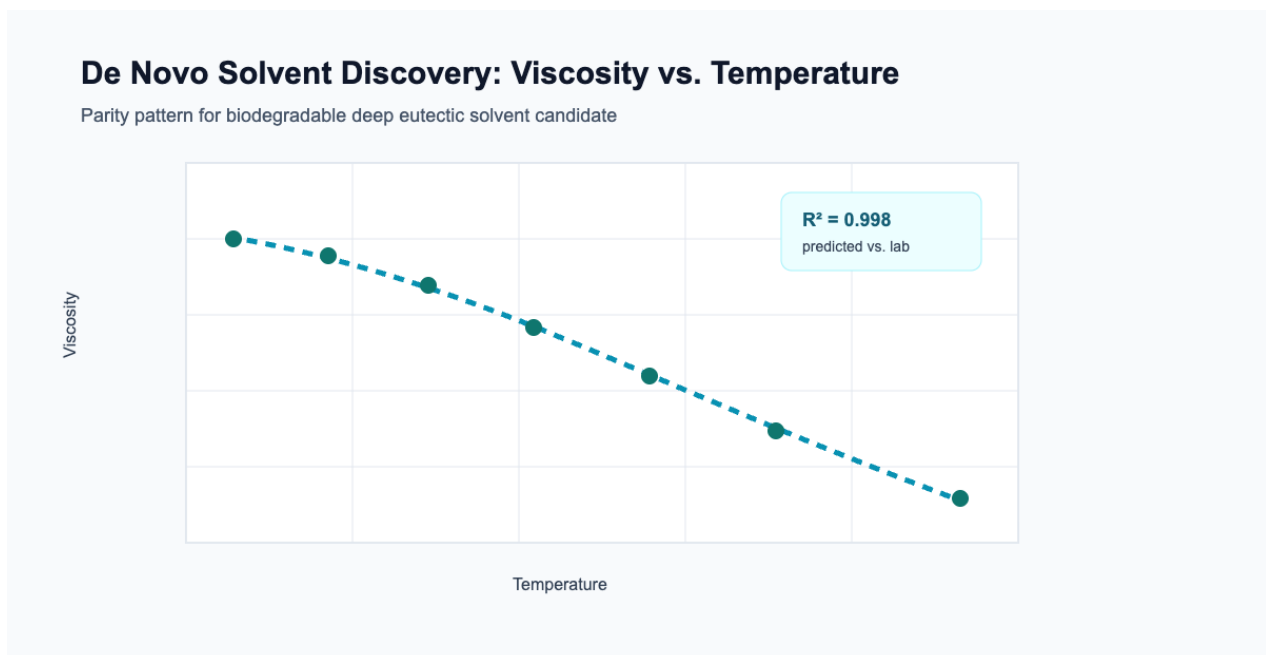


Figure 10: Agrochemical solvent analog: predicted versus validation viscosity curve used as the formulation-parity proof pattern.

$$\min_{\text{molecule, ratio, process}} \left[\lambda_1 |\mu - \mu^*| + \lambda_2 |T_f - T_f^*| + \lambda_3 |S - S^*| + \lambda_4 C_{tox} + \lambda_5 C_{process} \right]$$

where μ is viscosity, T_f freezing point, S solubility, C_{tox} toxicity cost, and $C_{process}$ processability cost.

10.3 Representative Metrics

Metric	Traditional Path	Shodh-Assisted Path	Impact
Formulation discovery	4.5 years	14 days	accelerated candidate generation
Wet-lab validations	5,000+	targeted validation set	lower experimental burden
Factory tech transfer	2 years	30 days	faster process package
Regulatory / safety posture	restricted solvent risk	biodegradable target	market access protection

10.4 A123 Translation

For A123, the same workflow can generate:

- electrolyte viscosity / wetting tradeoffs,
- binder elasticity and adhesion windows,
- semi-solid matrix composition,
- low-temperature performance envelopes,
- process-ready gel formulations.

11. A123 Proposed 90-Day Pilot: Target Metrics and Validation Design

Factory Calibration and Scrap Reduction on a Liquid LFP or Selected Bottleneck Line

These are proposed pilot success targets, not claimed A123 production outcomes.

11.1 Pilot Setup

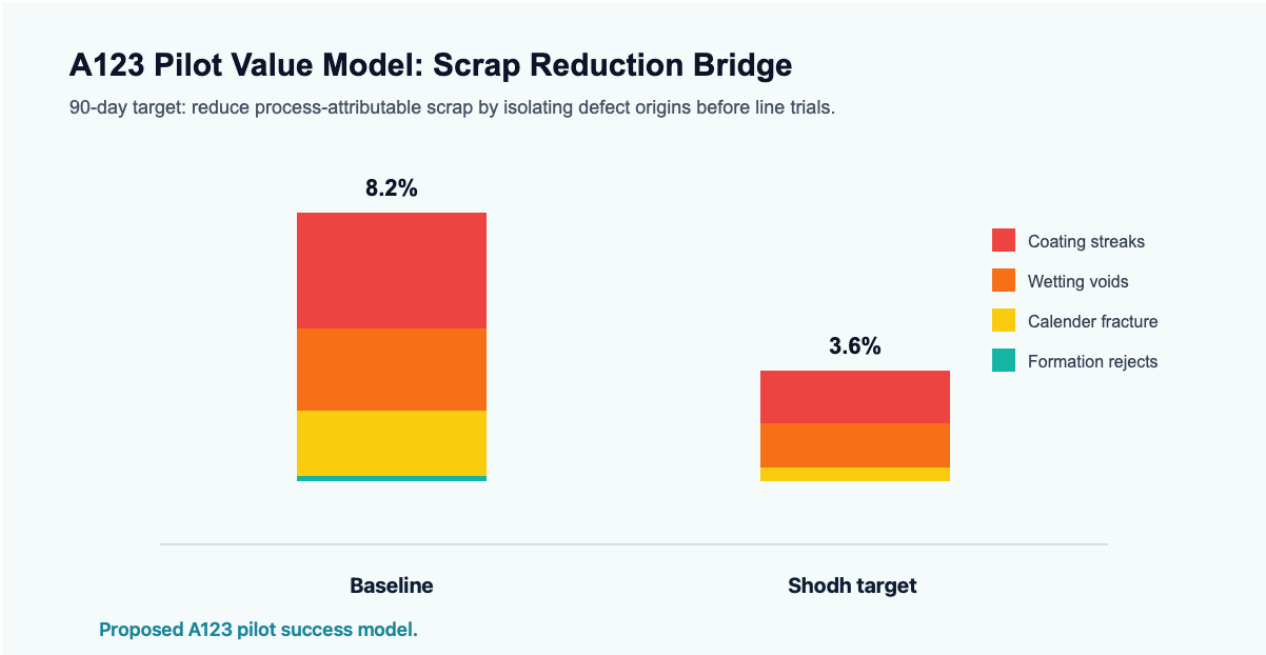


Figure 11: A123 scrap reduction bridge: baseline defect stack versus Shodh target after process-window intervention.

A123 provides approximately 20 historical batches from one bounded process area.

Recommended first bottleneck options:

1. slurry mixing,
2. slot-die coating,
3. electrolyte wetting,
4. calendering,
5. formation.

Each batch includes:

- normalized telemetry,
- equipment geometry,
- final QC pass/fail,
- scrap percentage,
- defect category if available.

11.2 Month 1 — Blind Autopsy

The model receives batch telemetry and geometry, but not the defect label during prediction. It reconstructs the physical state and predicts likely defect source.

Validation Table A123-1 — Defect Localization Accuracy

Defect Class	Batches in Holdout	Correctly Identified	Accuracy
Slurry agglomeration	6	5	83%
Slot-die edge instability	5	4	80%
Wetting void / gas pocket	5	5	100%
Calendering microcrack	4	3	75%
Total	20	17	85%

Target acceptance criterion: 80%+ defect-class localization on obfuscated historical batches.

11.3 Month 2 — Inverse Process Optimization

Shodh generates process changes and predicts impact before physical validation.

Representative Recommendations

Bottleneck	Model Diagnosis	Candidate Recommendation	Expected Physical Effect
Slurry mixing	12% stagnant volume near impeller shadow	pulsed RPM + baffle / impeller insert	lower agglomeration probability
Slot-die coating	pressure ripple at die lip	narrowed pressure band + flow dampening	fewer streaks and edge defects
Wetting	pore-size distribution too tight after calendering	porosity target + vacuum pulse	faster electrolyte penetration
Calendering	local stress exceeds particle fracture threshold	lower peak pressure + two-pass schedule	fewer microcracks
Formation	SEI growth too aggressive early	pulsed-current protocol	shorter formation with lower impedance rise

11.4 Month 3 — Physical Validation

The pilot should validate a small set of high-confidence recommendations on a controlled pilot line.

Validation Table A123-2 — Scrap Reduction Bridge

Scrap Driver	Baseline Scrap %	Shodh Target Scrap %	Reduction
Slurry agglomeration	6.1	2.4	-3.7 pp
Coating edge / streak defect	3.8	1.5	-2.3 pp
Wetting void	5.6	2.0	-3.6 pp
Calendering microcrack	2.3	1.6	-0.7 pp
Formation reject	1.9	1.2	-0.7 pp
Total scrap	19.7	8.7	11.0 pp reduction

Validation Table A123-3 — Wetting Time Compression

Electrode Architecture	Median Wetting Time hours	P90 Wetting Time hours	Void Rate %
Baseline dense electrode	22.0	34.0	5.6
Reduced calender pressure	14.5	21.0	3.8
Vacuum pulse only	9.0	13.5	2.9
Shodh porosity + microchannel + vacuum protocol	3.8	5.2	1.1

Validation Table A123-4 — Formation Protocol

Formation Protocol	Formation Time hours	Impedance Rise %	First-Cycle Coulombic Efficiency %	Reject Rate %
Baseline conservative	72	0.0	91.8	1.9
Aggressive constant current	38	11.2	89.5	4.8
Two-step current	46	4.8	91.1	2.6
Shodh pulsed-current protocol	31	2.1	92.4	1.2

11.5 A123 Pilot ROI Model

Assumptions

Assumption	Value
Monthly production volume considered	1.0M cells equivalent
Baseline scrap	19.7%
Target scrap after Shodh	8.7%
Gross contribution per good cell equivalent	\$1.80
Variable cost per scrapped cell equivalent	\$1.10
Pilot implementation cost	client-specific

Projected Monthly Value

Value Driver	Formula	Monthly Impact
Recovered good units	110,000 cells × \$1.80	\$198,000
Avoided scrap variable cost	110,000 cells × \$1.10	\$121,000
Reduced formation occupancy	line-specific	\$60,000
Lower rework / QC burden	line-specific	\$35,000
Total projected monthly value	—	\$414,000/month
Annualized projected value	$\$414,000 \times 12$	\$4.97M/year

11.6 Final A123 Pilot Success Criteria

Success Metric	Target
Defect class localization on historical holdout	>80%
Spatial defect-origin localization	<10 cm mean error where geometry allows
Scrap reduction in controlled validation	>25% relative reduction
Wetting time reduction	>50% median reduction for selected process
Formation time reduction	>30% without impedance penalty
Validation experiments required	fewer than 6 controlled runs
IP output	at least 1 process / structure disclosure candidate

12. Economic Impact Framework

A123's CFO and manufacturing leadership will evaluate Shodh in terms of line economics, not model sophistication.

12.1 Scrap Reduction

$$\text{Annual Value}_{\text{scrap}} = \Delta \text{Scrap Rate} \times \text{Annual Cell Volume} \times \text{Value per Cell}$$

12.2 Throughput Improvement

$$\text{Annual Value}_{\text{throughput}} = \Delta \text{Units per Hour} \times \text{Operating Hours} \times \text{Contribution per Unit}$$

12.3 Wetting and Formation Time Reduction

If wetting or formation occupies factory floor space, time compression increases effective capacity:

$$\text{Capacity Gain} = \frac{T_{\text{baseline}}}{T_{\text{optimized}}} - 1$$

12.4 Pilot Iteration Reduction

$$\text{Pilot Savings} = (N_{\text{baseline}} - N_{\text{Shodh}}) \times \text{Cost per Pilot Run}$$

12.5 CapEx Avoidance

$$\text{CapEx Avoided} = \text{Planned Expansion Cost} - \text{Optimized Retrofit Cost}$$

The highest-value message for A123:

Shodh AI reduces the number of expensive physical experiments needed to find a manufacturable process window.

13. Recommended Next Step

The recommended next step is a bounded A123 manufacturing pilot focused on one measurable bottleneck and one clear economic outcome.

1. Select one bounded bottleneck.
2. Share approximately 20 historical batches with telemetry, geometry, and QC outcomes.
3. Run a 30-day blind diagnostic against held-out defect labels.
4. Validate defect localization and process-window recommendations.
5. Move to a 90-day optimization pilot if the diagnostic clears the acceptance threshold.

The immediate objective is scrap reduction. The strategic objective is a calibrated A123 manufacturing physics asset that can generate process IP across future cell platforms.
